# EXP NO: 18 EXPERIMENT TO UNDERSTAND MATPLOTLIB LIBRARY USE CASES IN DATA SCIENCE THROUGH VISUALIZATION

## AIM:

Use Iris data set to understand the Matplot lib library.

## PYTHON CODE:

```
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt

data = pd.read_csv('/content/Iris_Dataset.csv')
data.info()
data.describe()
data.value_counts('variety')
sns.countplot(x='variety', data=data)
plt.show()
dummies = pd.get_dummies(data.variety)

FinalDataset = pd.concat(
    [pd.get_dummies(data.variety), data.iloc[:, [0, 1, 2, 3]]],
    axis=1
)

FinalDataset.head()
sns.scatterplot(
    x='SepalLengthCm',
    y='SepalWidthCm',
    hue='variety',
    data=data
)
plt.show()
sns.scatterplot(
    x='PetalLengthCm',
    y='PetalWidthCm',
    hue='variety',
    data=data
)
plt.show()
```

```
sns.pairplot(data, hue='variety', height=3)
plt.show()

sns.FacetGrid(data, hue='variety', height=5).map(
   sns.histplot, 'PetalLengthCm'
).add_legend()
plt.show()

sns.FacetGrid(data, hue='variety', height=5).map(
   sns.histplot, 'PetalWidthCm'
).add_legend()
plt.show()

sns.FacetGrid(data, hue='variety', height=5).map(
   sns.histplot, 'SepalLengthCm'
).add_legend()
plt.show()

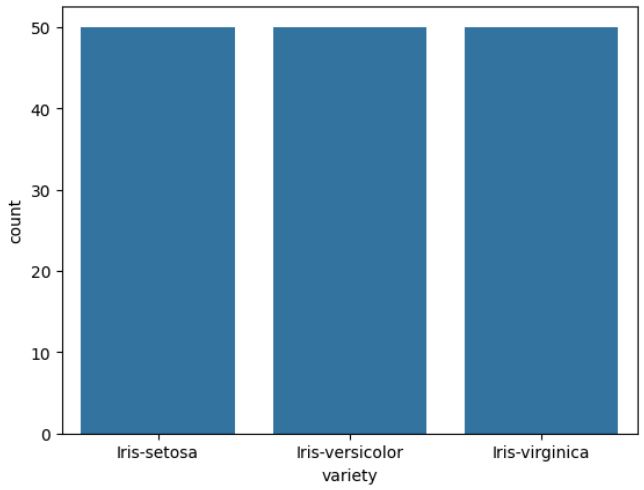
sns.FacetGrid(data, hue='variety', height=5).map(
   sns.histplot, 'SepalWidthCm'
).add_legend()
plt.show()
```

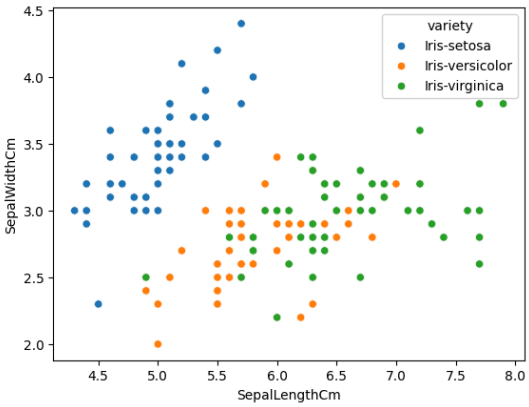
# OUTPUT:

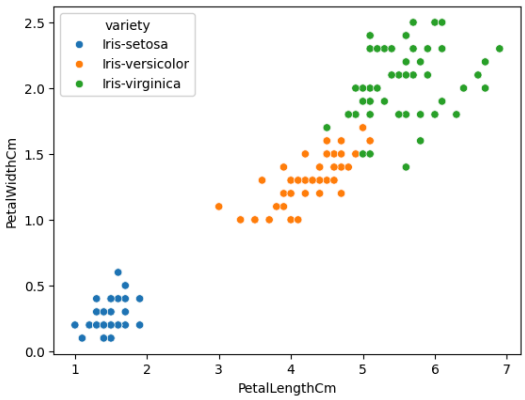
```
     SepalLengthCm  SepalWidthCm  PetalLengthCm  PetalWidthCm   Id  \
0              5.1           3.5            1.4           0.2    1
1              4.9           3.0            1.4           0.2    2
2              4.7           3.2            1.3           0.2    3
3              4.6           3.1            1.5           0.2    4
4              5.0           3.6            1.4           0.2    5
..             ...           ...            ...           ...  ...
145            6.7           3.0            5.2           2.3  146
146            6.3           2.5            5.0           1.9  147
147            6.5           3.0            5.2           2.0  148
148            6.2           3.4            5.4           2.3  149
149            5.9           3.0            5.1           1.8  150

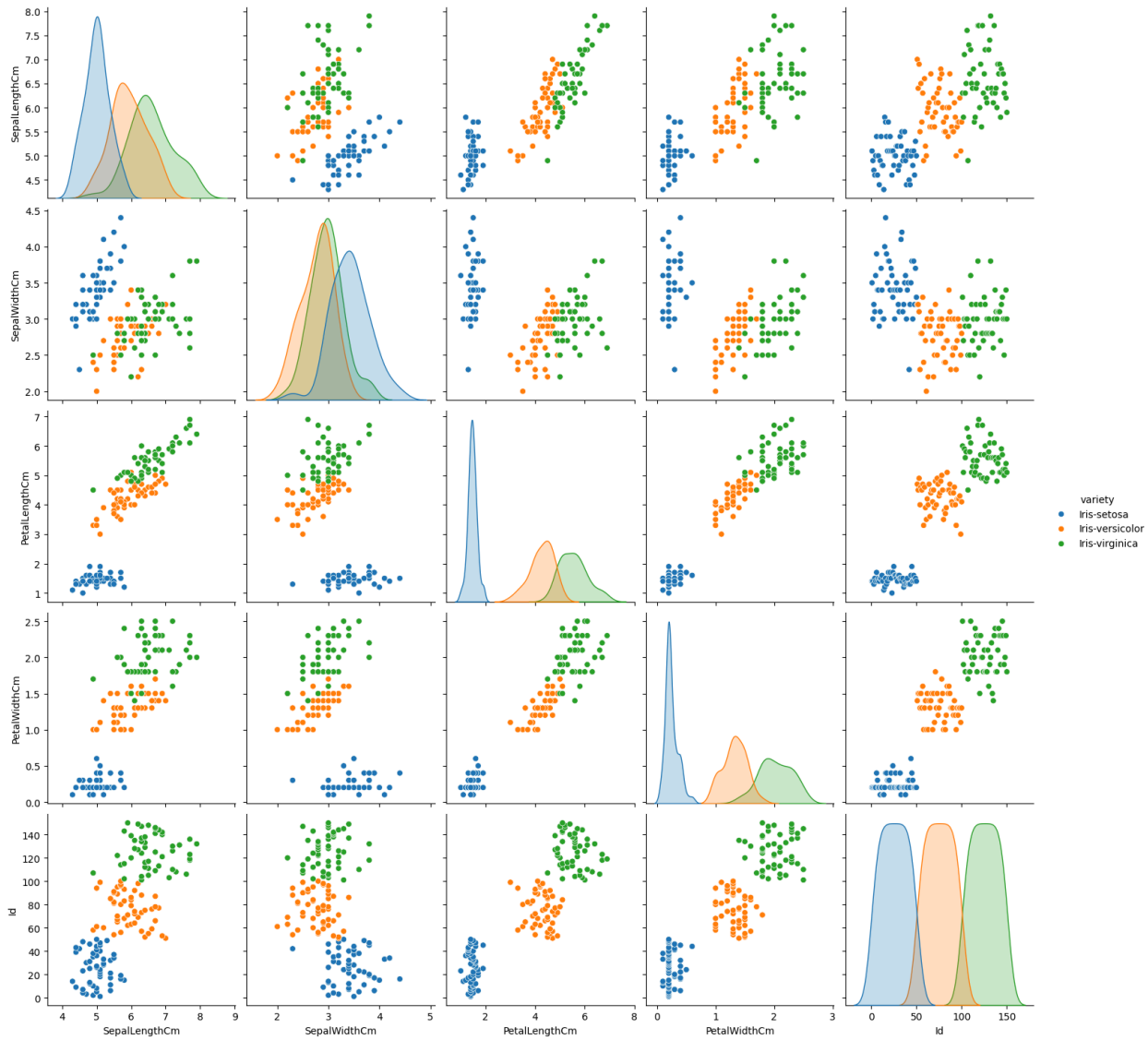
            variety
0       Iris-setosa
1       Iris-setosa
2       Iris-setosa
3       Iris-setosa
4       Iris-setosa
..              ...
145  Iris-virginica
146  Iris-virginica
147  Iris-virginica
148  Iris-virginica
149  Iris-virginica

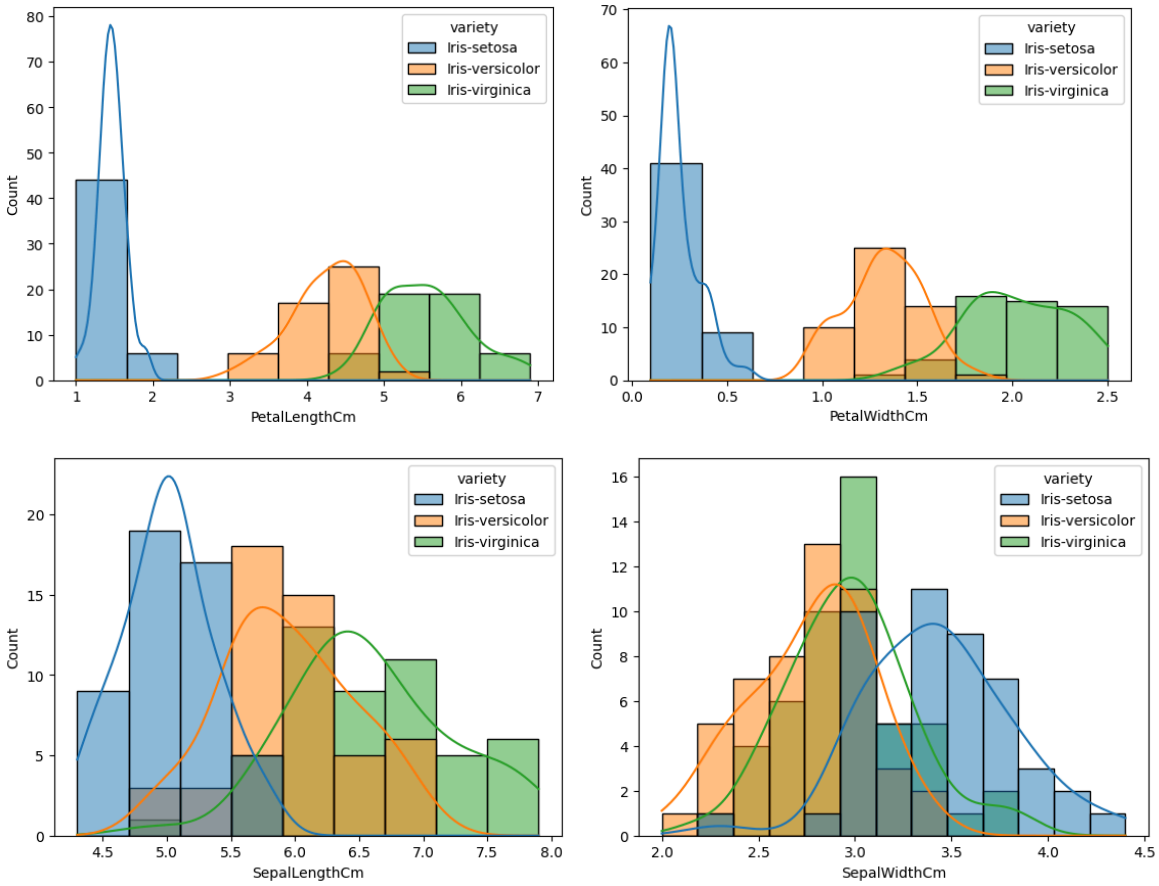
[150 rows x 6 columns]
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 6 columns):
 #   Column         Non-Null Count  Dtype
---  ------         --------------  -----
 0   SepalLengthCm  150 non-null    float64
 1   SepalWidthCm   150 non-null    float64
 2   PetalLengthCm  150 non-null    float64
 3   PetalWidthCm   150 non-null    float64
 4   Id             150 non-null    int64
 5   variety        150 non-null    category
dtypes: category(1), float64(4), int64(1)
memory usage: 6.3 KB
       SepalLengthCm  SepalWidthCm  PetalLengthCm  PetalWidthCm          Id
count     150.000000    150.000000     150.000000    150.000000  150.000000
mean        5.843333      3.057333       3.758000      1.199333   75.500000
std         0.828066      0.435866       1.765298      0.762238   43.445368
min         4.300000      2.000000       1.000000      0.100000    1.000000
25%         5.100000      2.800000       1.600000      0.300000   38.250000
50%         5.800000      3.000000       4.350000      1.300000   75.500000
75%         6.400000      3.300000       5.100000      1.800000  112.750000
max         7.900000      4.400000       6.900000      2.500000  150.000000
variety
Iris-setosa        50
Iris-versicolor    50
Iris-virginica     50
Name: count, dtype: int64
```




variety
Iris-setosa
Iris-versicolor
Iris-virginica
PetalWidthCm
PetalLengthCm


SepalLengthCm
SepalWidthCm
PetalLengthCm
PetalWidthCm
Id
variety
Iris-setosa
Iris-versicolor
Iris-virginica




## RESULT:

Thus the code is executed successfully.